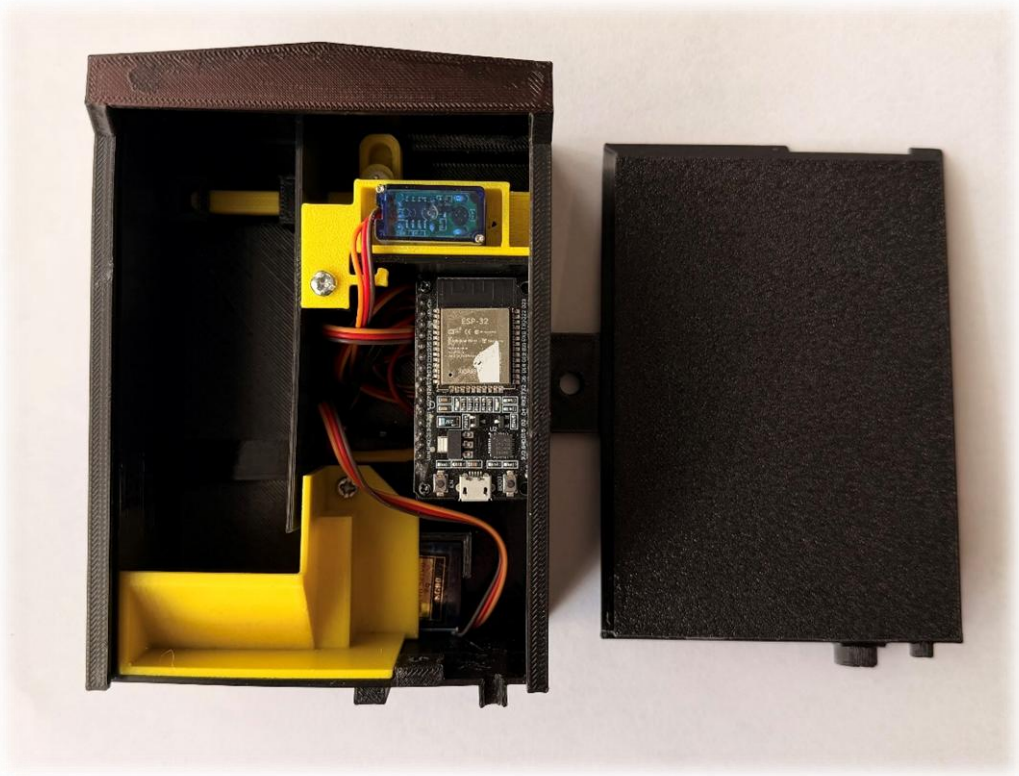


Wireless Automated Key-Release System

Assembly Guide



Designed By: Bryan Belcher

Contents

Required Components:.....	3
Additional Notes:	3
Demonstration Video:	3
Wiring Diagram:	4
Controlling Command:	4
Step 1:	5
Step 2:	5
Step 3:	6
Step 4:	6
Step 5:	7
Step 6:	8
Step 7:	8
Step 8:	9
Step 9:	9
Step 10:	10

Required Components:

1. Door x 1 [3D Printed]
2. GuideSleeve x 1 [3D Printed]
3. KeyDropEnclosure x 1[3D Printed]
4. Lid x 1 [3D Printed]
5. LockPin x 1 [3D Printed]
6. MotorSleeve x 1 [3D Printed]
7. PinKnob x 1 [3D Printed]
8. ESP32 Micro Controller x 1
9. ESP32 Power Cable x 1
10. Micro Servo Motor SG90 x 2
11. M1.7*5 Screw x 6
12. #6 x 3/8" Screws x 2

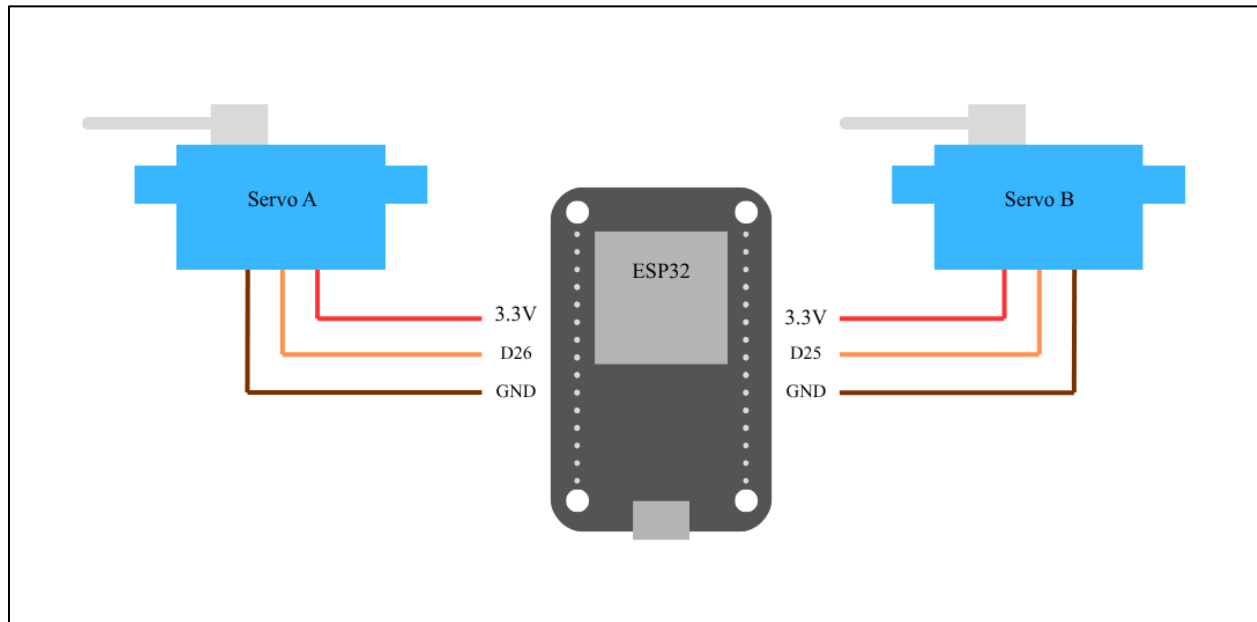
Additional Notes:

- Print all 3D printed components in a high-temperature capable filament. For this I am using Creality PETG.
- Slight screw size variation may be allowed as material is plastic.
- If the assembly is installed in an area exposed to weather, applying a resin coat to the top surface can help protect it.

Demonstration Video:

YouTube Link

Wiring Diagram:



Controlling Command:

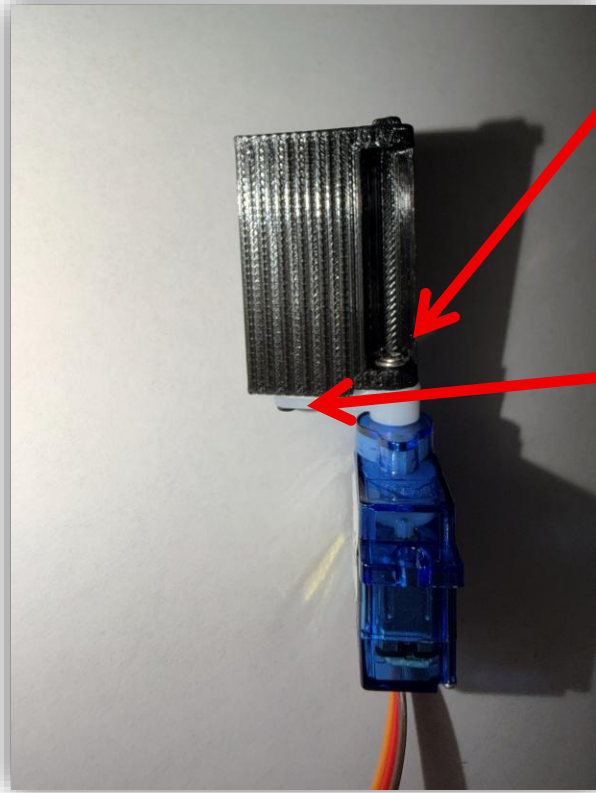
`http://[Local IP address]/[function]?key=[Password]`

Example:

`http://192.168.01.123/drop?key=p@ssw0rd`

Step 1:

Attach the **[Door]** to **Servo A** using the servo included screws.

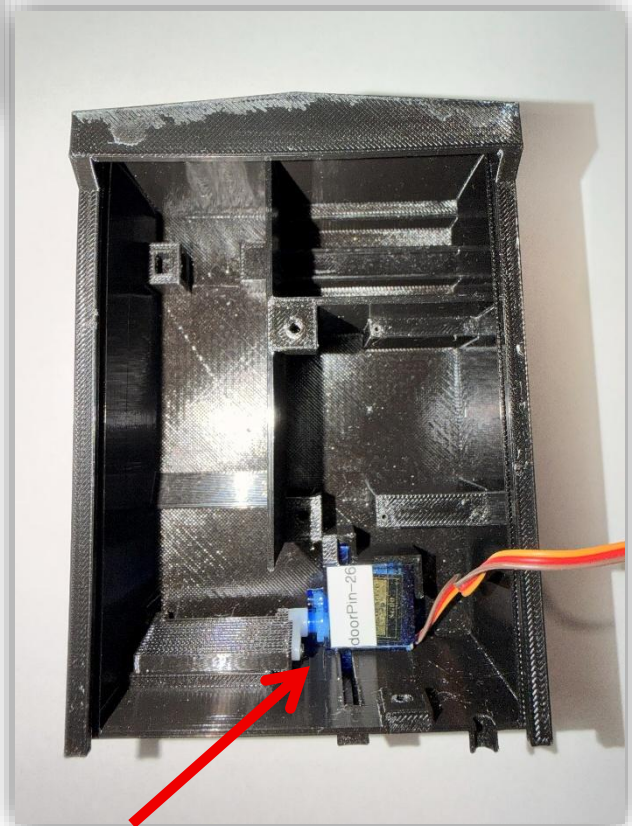


Servo included screw

Servo included screw

Step 2:

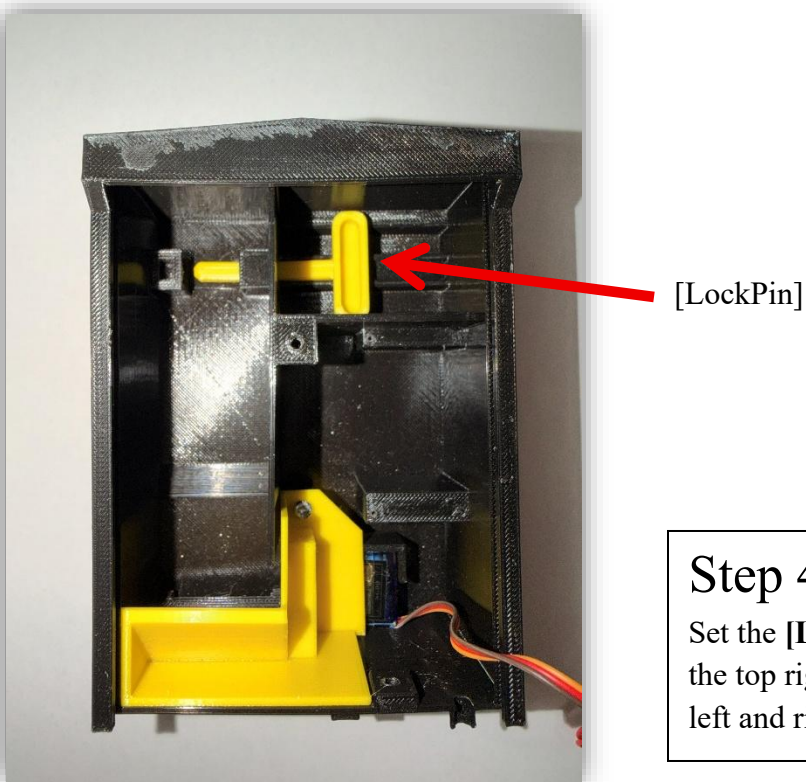
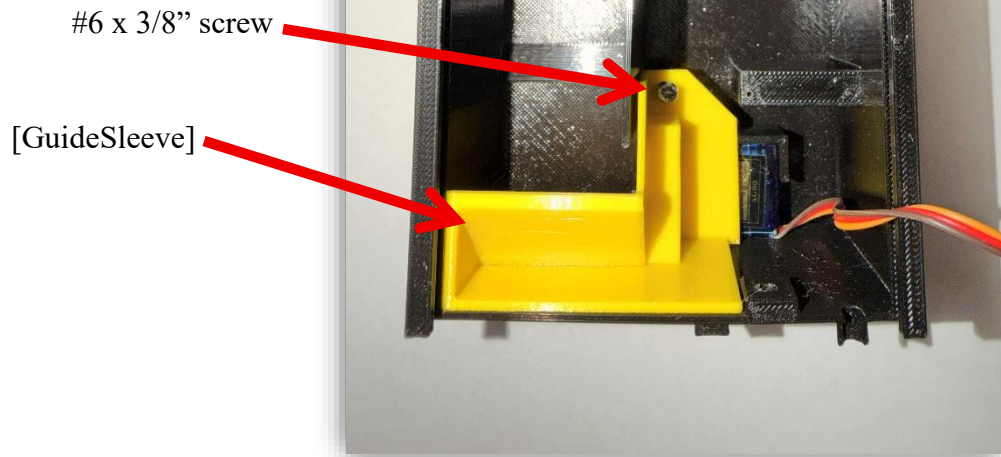
Slide **Servo A** in the slotted grooves on the bottom section of the **[KeyDropEnclosure]** so that it fits snug.



Servo with **[Door]**

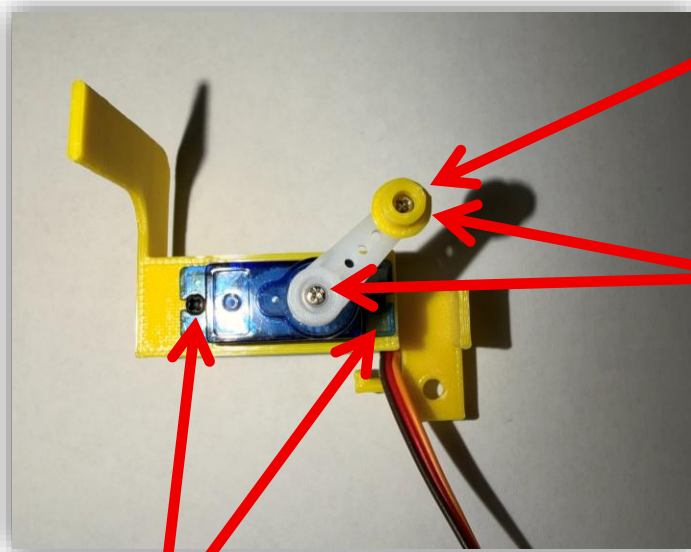
Step 3:

Set the **[GuideSleeve]** in place where it comfortably wraps around the installed **Servo A** and **[Door]** parts. Align the screw holes and thread in a #6 x 3/8" screw.



Step 4:

Set the **[LockPin]** in the slotted rail from the top right side so that it can freely slide left and right.



[KnobPin]

Servo included screws

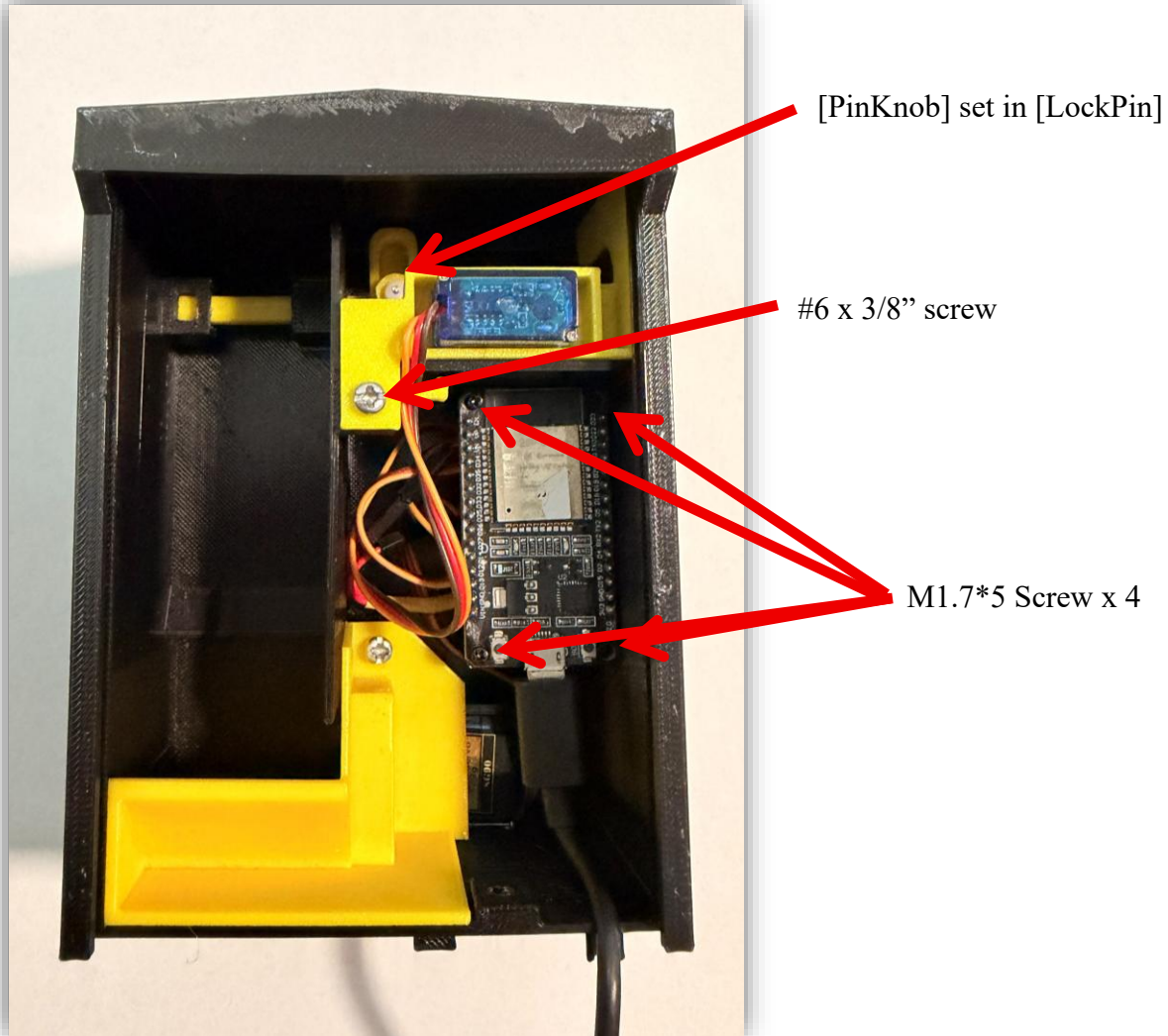
M1.7*5 Screw x 2

Step 5:

Attach the **[PinKnob]** to the included servo arm and the servo arm to **Servo B** using the included screws. Place the servo in the **[MotorSleeve]** and attach using M1.7*5 Screw x 2.

Step 6:

Set **Servo B** and **[MotorSleeve]** into the top right opening with the **[PinKnob]** sitting in the **[LockPin]** groove. Screw the **[MotorSleeve]** in place using a #6 x 3/8" screw.



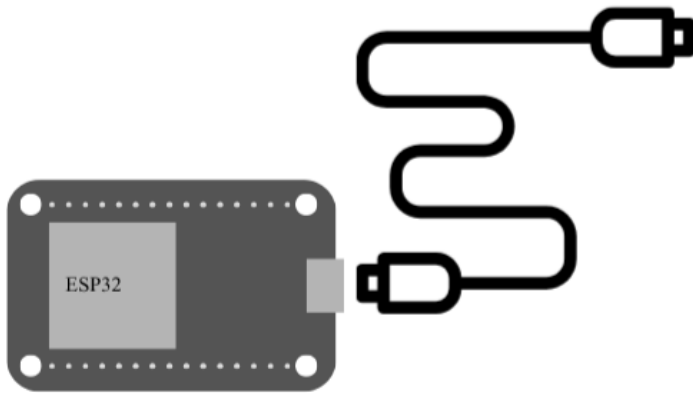
Step 7:

Wire the two servo motors to the **ESP32** controller according to the **Wiring Diagram**. Screw the **ESP32** controller to the **[KeyDropEnclosure]** using M1.7*5 Screw x 4.

Step 8:

Connect the **ESP32** and a computer using the data transferable **Power Cable** and upload the GitHub provided project program to the **ESP32**.

Adjustments to the program file is required to allow network access. Servo angle changes in program file may also be necessary. Arduino IDE is a great tool for this.



Step 9:

Slide the [Lid] over the assembly, then secure it by driving a #6 x 3/8" screw into the designated screw hole.



M1.7*5 Screw x 1

Step 10:

After statically assigning the IP address in your router's settings, activate the mechanism by the running the **Controlling Command** from a device on the local network.

Examples:

